Assignment 1 fall 25

[Only answers are provided here. You have to show the steps].

(a) $(119.015625)_{10} = (1110010.000001)_2$

(b) $(5963)_7 = (1956)_{10} = (B76)_{13}$

(c) $(593.0618)_{10} = (1001\ 1101\ 0111 . 0100\ 1010\ 0101\ 1100)_{Excess4}$

(d) $(110\ 1010\ 1010\ 1111\ 0101 . 0111\ 0101\ 011)_2 = (6AAF5.756)_{16}$

(e) $(1011010110)_{1's} = -(0100101001)_2 = -(601)_{10}$

(f) $(0011011101)_{2's} = +(0011011101)_2 = +(445)_{10}$

(g) $(4463.0712)_8$

$= (100100110011.000111001010)_2$

(h) $(1100\ 0011\ 0110\ 1011\ 1010 . 0101\ 1001)_{Excess3}$

$= (90387.26)_{10}$

(i) $(1011010110)_{2's} = -(0100101010)_2 = -(602)_{10}$

(j) $(67)_{10} = (151)_6$

(k) $(100110.1101)_4 = (1044.31640625)_{10} = (1380.27558)_9$

Question 2

$(2FB4)_{18}$

$+\ (8B)_{18}$

$(2G1F)_{18}$

Base-18

```
  2FB4
-   8B
------
  2F2B
```

Base 18

```
    2FB4
  ×   8B
--------
   1D9F8   [show working]
  14GHEX
--------
  16C9B8
```

Question 3

Base-6

```
112 | 101224 | 000503
    | 0
    ------
      10
       0
      ---
      101
        0
      ---
      1012
      1004
      ----
        42
         0
      ----
        424
        340
      ----
         44
```

112×1 = 112

112×2 = 224

112×3 = 340

112×4 = 452

112×5 = 1004

Quotient $(503)_6$

Remainder $(44)_6$

Question 6

$+(91)_{10} = +(1011011)_2$

$= +(01011011)_2$

$= (01011011)_{2's}$

$-35 = -(100011)_2$

$= -(00100011)_2$

$= (11011101)_{2's}$

```
   11 1111
   01011011
+  11011101
-----------
  100111000
  ×
```

$(00111000)_{2's}$

Adding two different sign numbers. So overflow not possible.